



Microalgal membrane fluidity monitoring using DCVJ

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ABSTRACT

This article proposes a flow cytometric methodology to monitor microalgae membrane fluidity using DCVJ (9-(2,2-DicyanoVinyl)Julolidine). DCVJ is a methanol-soluble molecular rotor that inserts itself into the cell membrane. The tagging procedure was optimized in terms of cell-to-probe ratio, probe-to-water ratio, incubation time, and methanol toxicity. The model microalgae for this work was *Chlorella vulgaris*. The results demonstrated that the best incubation conditions are 1 mL sample at 1 MCell/mL, with a probe addition of 10 μ L at 10 mM, incubated in the dark for 20 min. In addition, the protocol shows some robustness as cell concentration can be decreased to 0.1 MCell/mL, incubation time extended from 20 to 30 min, and methanol quantity quadrupled without perturbing the signal. Once optimized, the protocol was applied to monitor microalgae membrane acclimation to hyperosmotic (rigidifying) conditions and their reversion (fluidizing). An advanced signal treatment is required to access membrane fluidity properly. Ultimately, the protocol allowed monitoring for two passages, delivering explainable and repeatable results.

1. Introduction

Born in the depths of the oceans eons ago, microalgae have long left their marine cradle to spread all over the Earth. Owing to their incredible adaption capabilities, they can nowadays be encountered in places as different as deserts [1], volcanic water sources [2], or even glaciers [3]. Yet, not only did they spread over the entire world, they shaped the world we know. Their two main feats might be the production of the oxygen present in our atmosphere (about half of the current annual oxygen production [4]) and the support they offer the higher levels of the aquatic food webs [5]. Still, their promises extend far beyond what they already delivered. In the eyes of the biotechnologists, they appear as a tool to shift society toward a bio-based, sustainable future [6]. In the eyes of the ecologists, their sensibility to their immediate environment makes them valuable advanced bio-sensors to document the tremendous strain we are putting our ecosystems through [7,8].

This ability of the microalgae to react to their habitat condition can be taken advantage of at the photosynthetic apparatus level. Indeed, this very sensible piece of the cell's energetic metabolisms reacts to various stressors, even at low concentrations (e.g., microplastics [9], heavy metals [10–12]). The quantities of interest are classically the photosystem II quantum yield (a.k.a. F_v/F_m measurements) and the dynamic of the photosystem II fluorescence (a.k.a. OJIP tests). The former is a broad indicator of the photosynthetic apparatus general

well-being, while the latter allows for pinpointing the alterations of the electron flow with the electron transport chain. Another element that may indicate how microalgae cope with stress is the state of their membrane. Indeed, similarly to any prokaryotic or eukaryotic cell (archaea have a different membrane structure), microalgae membranes are phospholipid bi-layers that play a crucial role in protecting the cell from its environment. Membranes control water-soluble molecules trafficking in and out of the cells with the help of embedded proteins. Still, the bi-layer equilibrium is both paramount and subtle, as transitioning from the liquid-crystalline state (F_β) to the gel state (F_α) can lead to protein expulsion and, ultimately, cell death. Maintaining adequate fluidity is, therefore, of crucial importance for the cells.

The mechanisms associated with fluidity control are designated as homeoviscous. They encompass responses to various stresses, the most common one being heat and extracellular osmolarity variations. Indeed, a too-cold temperature, or a too-high osmolarity, induces a reduction of the membrane fluidity and can shift it to its gel state [13,14]. Still, many other extracellular factors can affect membrane fluidity. The most known to the microbiologist might be the presence of ethanol, which penetrates the bi-layer and destabilizes it [14,15]. To counter these adverse effects, cells alter the level of unsaturation of the acyl group in the bi-layer (higher unsaturation leads to a higher fluidity). They also produce sterols (cholesterol and ergosterol) to rigidify the membrane or oleate to fluidize it. Stabilization can also be led on the hydrophilic

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side of the bi-layer with the production of sugar (sucrose and trehalose) to maintain the distance between the hydrophilic heads [14]. Therefore, extracting and quantifying these compounds helps derive insights into how cells react to an adverse environmental change. Still, qualifying membrane components is only a very indirect way to assess its fluidity and can sometimes be misleading. For example, Da Silva et al. showed that ethanol tolerance in the bacterium *Oenococcus oeni* coincided with an increase of the acyl groups unsaturation of the membrane phospholipids. This is counterintuitive as this mechanism would further fluidize the membrane and worsen the impact of ethanol. Still, overall, when directly measuring membrane fluidity, a rigidification is observed [15]. Ultimately, it is a membrane-stabilizing protein that drives the central part of the membrane homeoviscous acclimation. This example underlines that direct fluidity measurement is not replaceable by membrane lipid composition analysis. Going one step further, it casts some serious doubts on protocols based on total fatty acid profiling (including storage lipids) to derive information on membrane state [16].

Still, direct measurement of membrane fluidity can be particularly challenging. Indeed, like any membrane-related measurement, fluidity assessment requires introducing a probe within the bi-layer and recovering a form of signature [17]. This is classically done by placing cells in contact with DPH (1,6-DiPhenyl-1,3,5-Hexatriene, a highly hydrophobic molecule finding refuge in the cell membrane). Once trapped in the membrane, DPH can rotate to a certain extent, determined by the local fluidity. Therefore, if one excites it with polarized light and recovers the polarization of its fluorescence, he can compute the level of anisotropy, which relates to DPH rotation, hence local fluidity [13] (the higher the anisotropy, the lower the fluidity). In addition to its conceptual complexity, this method requires specific devices and usually probes a pool of cells [18]. This last drawback might be the most problematic one. Indeed, when exposed to stress, not all the cells within a population react in the same manner. For example, oxidative stress exposure, which turns *Haematococcus pluvialis* red, also induces a high mortality [19]. Therefore, distinguishing between the two populations is important in order to identify the dynamic of each of them precisely.

Probing cells individually is a challenge solved by medicine researchers a few decades ago thanks to a technique referred to as flow cytometry. Flow cytometry is the conjunction of two technologies: hydrodynamic focusing and optoelectronic. Hydrodynamic focusing is the manipulation of the flow of a suspension so that particles (or cells in this case) can be set in line and passed through a laser beam one at a time, and at a high frequency (hundreds of events per second). This method solves the first part of the problem: individualizing cells. Optoelectronic then handles the second: probing the cell. Within the laser beam, cells interact with light in two manners: particulate and molecular. Particulate interaction leads to Front Scatter (FSC signal, related to size) and Side Scatter (SSC, related to morphology). Molecular interaction originates from endogenous and exogenous molecules, capturing the laser light and emitting it as fluorescence. In both cases, light is collected, amplified, and sorted spectrally. Finally, the recovered signal intensity is interpreted based on the involved molecules' mode of action and/or metabolic relevance.

Owing to its flexibility, flow cytometry has been used to address a broad spectrum of challenges in microalgal ecotoxicology and biotechnology. Notable applications include morphology monitoring [20], quantification of Reactive Oxygen Species (ROS) levels [21], viability assessments [22], tracking changes in cell lipid content under stress [23], analyzing marine population composition [24], monitoring ROS-induced pigment production [19], selecting high-lipid producing mutants [25], and, recently cell division tracking [26].

Surprisingly, less than a handful of teams used flow cytometry to assess membrane fluidity at the cell scale. Among them, Fox et al. introduced a complex modification of an EPICS V flow cytometer (addition of beam splitters, and horizontal and vertical polarizers) to detect DPH

emission polarization [27]. While promising, it is only thirty years later that the technique reemerged in an article by Bouix et al. [28]. Given the level of technicality of the method and the expertise held by the two teams, one can easily understand why the method did not spread widely and is unlikely to in the future. The last team who mentioned flow cytometry monitoring of membrane fluidity is Gallardo-Rodriguez et al. who investigated shear-induced membrane alteration in *Protoceratium reticulatum* (dinoflagellate) [29]. Interestingly, they relied on another probe, 9-(2,2-DiCyanoVinyl)Julolidine (DCVJ), which might be easier to handle as a regular, blue-laser mounted, unmodified, flow cytometer can detect its signal. Unfortunately, the protocol is described in a somewhat expeditive manner, seriously hindering its reproducibility. Fortunately, guidance can be recovered from a biomedical study having employed DCVJ to study membrane fluidity in endothelial cells under the conditions of fluid shear stress [30]. Where authors used fluorophotospectrometry to ascertain a linear dose-response relationship between shear stress and membrane fluidity changes.

Taking a step back, it is interesting to learn more about DCVJ. This molecule was introduced, in details, by Haidekker et al. as a molecular rotor to probe local viscosity in fluid (oil) flow using fluorescence [31]. The molecule is composed of a julolidine group, which acts as the anchor, and a dicyanovinyl group, which acts as the rotor. When excited by a laser, the molecule tends to dissipate excess energy by rotation of its dicyanovinyl group; fluorescence is, therefore, low. Yet, when placed in a viscous environment, the rotor motion is restricted, and the molecule de-excitation takes place through the emission of green radiation (475 to 575 nm in ethylene glycol/glycerol mixture) by the julolidine group. Consequently, the probe offers a direct measure of local viscosity. In the context of membrane fluidity measurement, the probe is easily dissolved in methanol before being added to the cell suspension. Once in the cell suspension (aqueous medium), it will promptly enter the membrane bi-layer to flee from the polar environment.

Still, our first trials with this molecule taught us that things were not as easy as they seemed. In addition to classical protocol implementation and optimization, the probe signal is to be handled with care as the presence of DCVJ within the membrane alters its properties. To circumvent the problem, we conceptualized a mechanistic model of the probe action and formulated it into equations, allowing us to recover membrane fluidity information. Consequently, this article shares the tagging procedure we developed and optimized. It also clearly explains how to handle the generated data and why. Finally, the overall protocol is validated by monitoring *Chlorella vulgaris* membrane fluidity evolution over the course of two consecutive hyperosmotic shocks.

2. Materials and methods

2.1. Strain and culture medium

Chlorella vulgaris (CV 211-11b, SAG Culture Collection, Germany) was the model microalgae for this work. Double-concentrated Bold's Basal Medium [32] with tripled nitrogen content (classically referred to as B3N) and 20 g/l of glucose was used throughout this study (mineral and glucose solutions sterilized by autoclaving, separately). Heterotrophy was chosen as it allows for the safe triggering of hyperosmotic shocks (through high glucose concentration; see hereinafter for details on the protocol validation run). Furthermore, setup (strain, medium, mode of cultivation) can be considered an expansion of the application DCVJ to green microalgae, heterotrophy, low pH (6.2–6.5), and low osmolarity (24 to 135 mOsm/L) compared to the work of Gallardo-Rodriguez et al. (dinoflagellate, photoautotrophy, 8.0–8.2, above 1000 mOsm/L) [29].

Cells were cultivated in flasks (250 mL, 50 mL medium, orbital shaking 100 rpm) at a temperature of 15 °C. To avoid any artifact originating from medium acclimation, cells were subcultured numerous times (five or more) before being used for the tests.

For each sample, *Chlorella vulgaris* cells were counted to obtain cell concentration and adjust accordingly (hemocytometer counting, Zeiss Axioplan 2).

2.2. DCVJ and stock solution

DCVJ (9-(2,2-DiCyanoVinyl)Julolidine) was bought from Fischer Scientific (brand: Medchem Express). DCVJ was dissolved in pure methanol to reach a 10 mM (2.5 mg/ml) concentration and placed in a dark vial before storage at -20°C . One should note that DCVJ is difficult to weigh (high static sensibility). Therefore, we opted to add the corresponding volume of methanol in the vial into which the product was delivered (e.g., 4 ml methanol for a 10-mg delivery). This way, it was possible to recover the compound easily and entirely.

2.3. DCVJ tagging procedure & optimization

Adequate DCVJ tagging of *Chlorella vulgaris* cells should demonstrate three characteristics. The first is a good contrast between the tagged and control cells to allow for a clear separation. In terms of quantitative indicators, the higher the distance between the two populations, the better. Then, the signal recovered from tagged cells should exhibit a narrow distribution. This property can be measured as the coefficient of variation, ideally below 15% [33]. Finally, the tagging should be stable in time or at least offer a stability window wide enough (10 min or more) to allow comfortable measurement.

Numerous factors can influence the quality of the tagging. However, the two main ones are the cell-to-probe ratio and the incubation time. The cell-to-probe ratio can be evaluated using two protocols: keeping the probe concentration constant and varying the cell concentration, or the opposite. In our case, we tested both. Indeed, as DCVJ is a hydrophobic molecule, testing these two approaches allows for control for a third factor, namely, the water-to-probe ratio. The incubation time was evaluated from 0 (immediate analysis) to 120 min.

For these tests, we used the conditions identified during a preliminary testing phase: cell concentration of 1 MCell/mL, 10 μL of 10 mM DCVJ, 20 min of incubation in the dark at room temperature, and an extemporaneous analysis after incubation. As it will appear clear in the Results section, this procedure represents the optimum and will be referred to as the *reference* protocol hereinafter.

2.4. Maximal admissible methanol concentration

Methanol is known to induce cell membrane alteration. The most striking example might be the disruption-free extraction of chlorophyll when suspending microalgal cells in pure methanol. Therefore, it is important to verify that it does induce a detectable perturbation while delivering the DCVJ to the cell membranes. Thus, various cell-to-methanol ratios were tested. They ranged from 10:1 to 60:1 $\mu\text{L}:\text{mL}$, with a constant volume of initial *Chlorella vulgaris* suspension of 1 mL. As Canavate et al. pointed out, contact time matters [34]. In our case, it was set to 20 min as it is the optimum identified in the tagging optimization investigation. The probe fluorescence signal was monitored as the test outcome. From the proposed example (disruption-free pigment extraction), it can be expected that a too-high methanol concentration would decrease the probe signal as this molecule would fluidize the membrane and ultimately lead to pigment leakage.

2.5. Protocol validation run

The optimized protocol was tested by monitoring the membrane fluidity of a culture after its passaging. Indeed, cells are known to experience a reduction of the fluidity of their membranes when exposed to hyperosmolarity ([35,36] for detailed articles, [13,14] for reviews). From a practical point of view, cells were transferred from a mature culture (marginal glucose concentration, hence low/reduced osmolarity) to a fresh medium (with 20 g/L glucose, hence hyperosmolar). They were monitored approximately daily, depending on laboratory constraints and weekends. For each sample, cells were counted using a hemocytometer (Zeiss Axioplan 2) in order to follow their development

and adjust sample concentration for the tagging. The procedure was repeated (two passages monitored in a row) to confirm that the fluidity evolution originated from the passages. Finally, test cultures were conducted in biological triplicate.

2.6. Flow cytometry, data handling and statistical testing

Flow cytometry assays were performed using a BD Fortessa X20, operated with BD FACS Diva software. The analysis captured four key parameters: forward scatter (FSC, excited by a 488 nm blue laser) as an indicator of cell size; side scatter (SSC, also using a 488 nm laser with 488/10 nm detection) to estimate cell complexity or morphology; chlorophyll fluorescence (488 nm excitation, 695/40 nm emission); and DCVJ fluorescence (488 nm excitation, 530/30 nm emission). For each sample, a minimum of 100 000 events with FSC values above 5000 were recorded. Data processing was carried out using the FlowCal Python package. Cell populations were delineated based on their FSC and SSC profiles. To reduce background interference, chlorophyll fluorescence was employed, though background levels remained low. Extreme values were handled through two-sided 95% Winsorization [37], and all data were log-transformed (base 10) prior to statistical analysis.

Cohen's d index (Eq. (1)) was used as the statistical tool to analyze the similarity of flow cytometry readings. Indeed, in contrast with t -statistics and F -statistics, which lead to a p -value, Cohen's d is not asymptotically influenced by the number of samples. This point is important, as 100 000 events are recorded by analysis. Hence, comparing two analyses involves about 200 000 events, leading to extremely high t -statistics or F -statistics, zeroing any p -value. Conceptually, any observed difference is ascertained, no matter how small. In contrast, Cohen's d allows reporting a meaningful metric, provided enough events are included (which is the case here), showing how similar and close two populations are. It is calculated by norming the difference of the means of two populations by their pooled standard deviation [38,39]. It is, therefore, a dimensionless indicator. One can draw a general rule of thumb for cytometry readings: $d < 0.5$ represents a small difference, $0.5 < d < 1$ a medium one, $1 < d < 2$ a significant one, with the two populations starting to detach from one another, and $d > 2$ a clear indicator of two distinct populations [20]. As it is a pairwise indicator, it is possible to compute it for each couple of readings to be compared. Here, it was decided to avoid overcrowding the graphs by reporting pointless metrics. Hence, the focus was set on a comparison with a *reference* configuration.

$$d = \frac{\mu_1 - \mu_2}{\sqrt{\frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2}}} \quad (1)$$

Whence d is Cohen's metric, μ_1 and μ_2 the mean values of the two populations tested, σ_1 and σ_2 the standard deviations of the two populations tested, and n_1 and n_2 the number of samples of the two populations tested.

Apart from raw data analysis, the way signal intensity is related to fluidity is also to be presented. In their work, Haidekker et al. showed that the fluorescence intensity scaled to the power 0.59 with the viscosity (i.e., -0.59 for the fluidity) [31]. To determine this coefficient, the authors dissolved the probe in controlled ethylene glycol/glycerol mixtures and measured the DCVJ fluorescence. Hence, while it is difficult to get an exact value for the membrane fluidity, it is possible to access a relative one between two samples or for a whole experiment by defining a reference (e.g., the first point):

$$\Phi_r(t) = \left(\frac{F(t)}{F_{ref}} \right)^{-\frac{1}{0.59}} \quad (2)$$

Whence $\Phi_r(t)$ is the relative fluidity with respect to the reference at time t , $F(t)$ is the probe channel average signal at time t , and F_{ref} is the probe channel average signal at the reference time.

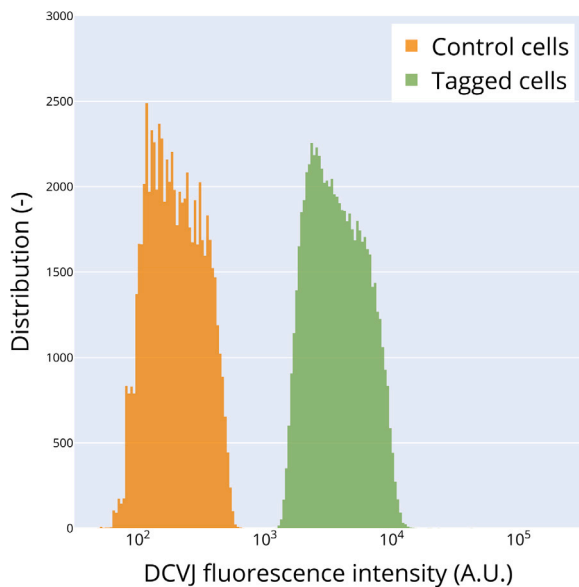


Fig. 1. Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) for the reference incubation conditions. Sample volume — 1 mL. Cell concentration — 1 MCell/mL. Probe volume and concentration — 10 μ L at 10 mM. Incubation time — 20 min in the dark.

3. Results and discussion

3.1. Tagging results

Fig. 1 presents the fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) for both control and tagged cells. As one can see, the two populations are well apart, with a distance of about 4120 fluorescence units (linear scale) or 1.28 decade (base 10 logarithm scale, used hereinafter as signal intensity are classically plotted on this type of analysis). This distance might be marginally increased by increasing the photomultiplier voltage. Yet, the expected gains are small, as the background signal of the control cell originates from their carotenoid content [40]. Furthermore, it would induce a risk of signal saturation, making the overall procedure unappealing. In terms of shape and standard deviations, the two populations exhibit a somewhat Gaussian shape (base 10 logarithm scale), with coefficients of variation of 9.77 and 6.46% for the control and tagged cells, respectively, which is admissible.

3.2. Optimization

Once control and tagged cells have been shown to be separated enough to allow for a proper distinction, the next step is to optimize tagging conditions. First, the effect of the probe-to-cell ratio was investigated. **Fig. 2** presents the effect of a varying cell concentration (Left), for a constant probe addition, and a varying probe concentration (Right), for a constant cell concentration. As one can see, cell concentrations of 0.1 and 1 MCell/mL yielded the best compromise between tagging intensity and population dispersion. On the contrary, a too-high cell concentration led to a decrease in the recovered signal intensity and a widening of the signal distribution. This effect can be attributed to the probable presence of multiple cells in the detection zone. In a different manner, a too-low concentration led to poor signal quality too. This time, the noise-to-signal ratio can be incriminated. Too few cells are present, and other events (background noise) widen the spread of the detected signal. Taking a step back, operating at 1 MCell/mL aligns well with classical flow cytometry operation guidelines and ensures fast analysis (a few minutes to collect 100 000 events). It is,

therefore, the optimum for this procedure. Focusing on the other way to manipulate the probe-to-cell ratio (varying probe concentration), one can see that probe concentrations of 20 and 40 mM yielded similar results as 10 mM. Diluting the probe by a factor of 2 or 5 also delivered admissible results (somewhat lower intensity but similar population spread). Still, further diluting the probe led to a decrease in tagging intensity and a widening of the signal distribution.

Following the probe-to-cell ratio, incubation time is to be optimized. **Fig. 3** — Left — reports the signal for control and tagged cells on the DCVJ fluorescence channel. From this graph, it can be seen that the probe signal (tagged cells) slowly rises between 0 and 30 min of incubation and abruptly decreases afterward before stabilizing. Over the process, the signal spread can be deemed constant. We can propose the following explanation for this behavior: inserting DCVJ in the membrane makes it more rigid and, consequently, triggers a slight homeoviscous reaction (oleate- or protein-related [14,15]) which requires about half an hour to come into play and fluidize the membrane. Therefore, if one desires to probe the original membrane state, he is to perform the measurement before these acclimation mechanisms alter the membrane state.

The control cells exhibit a different behavior. The background signal rises from 0 to 20 min before stabilizing. Its spread is also constant. The importance of the control cells will be discussed at length below. For now, it is sufficient to underline that a stable measurement can be obtained after 20 min. Similarly, stability of the control cells chlorophyll fluorescence signal importance will be stretched out. Yet, for now, it is enough to observe that it stabilizes after 15 min of incubation (**Fig. 3** - Right).

3.3. Maximal admissible methanol concentration

Fig. 4 presents the influence of increasing inclusion of methanol in the sample volume. The results show that the system is virtually undistributed for a volume of methanol of 50 μ L or less, after a 20 min exposure. Above, the recovered signal starts to decrease. This decrease was expected as high concentrations of methanol induce membrane leakage. From a practical point of view, it can be concluded that the proposed protocol (*i.e.*, adding 10 μ L of methanol carrying the probe to a 1-mL sample) is robust.

3.4. Protocol validation & advanced signal processing

The biological test unfolded as expected from a microbiology point of view. The cells grew twice in a row, and no contamination could be detected. In **Fig. 5** - Left -, the classical phases of a culture can be observed: lag, exponential, oxygen limited, and early stationary, in sequence, twice in a row. Focusing on the DCVJ reading (**Fig. 5** - Right -), they appear as quite confusing. Indeed, no clear trend emerges, and the use of Eq. (2) to compute the fluidity would not allow the recovery of information. For this observation, it is clear that optimal DCVJ tagging is not enough to analyze membrane fluidity over time.

In order to improve signal processing and extract meaningful information, additional elements have to be brought to the table. **Fig. 6** - Left — illustrates the concepts to be mobilized here. The correction process starts by remembering that adding molecules within a bi-layer membrane modifies it. This can be illustrated with the case of ethanol [15], as an exogenous stress, or of cholesterol/ergosterol and oleate [14] as homeoviscous strategies. Therefore, we can assume that DCVJ modifies the membrane it probes. One of the consequences of this alteration would be a modification of the fluorescence signal, on every channel (DCVJ, chlorophyll, or else). This claim is backed by the observations reported in **Fig. 3** - Right -, where the chlorophyll autofluorescence signal deviates from the control following DCVJ addition. Furthermore, there is no reason for the probe to alter the chlorophyll content *per se*, hence the signal modification can be linked to fluorescence properties with certainty.

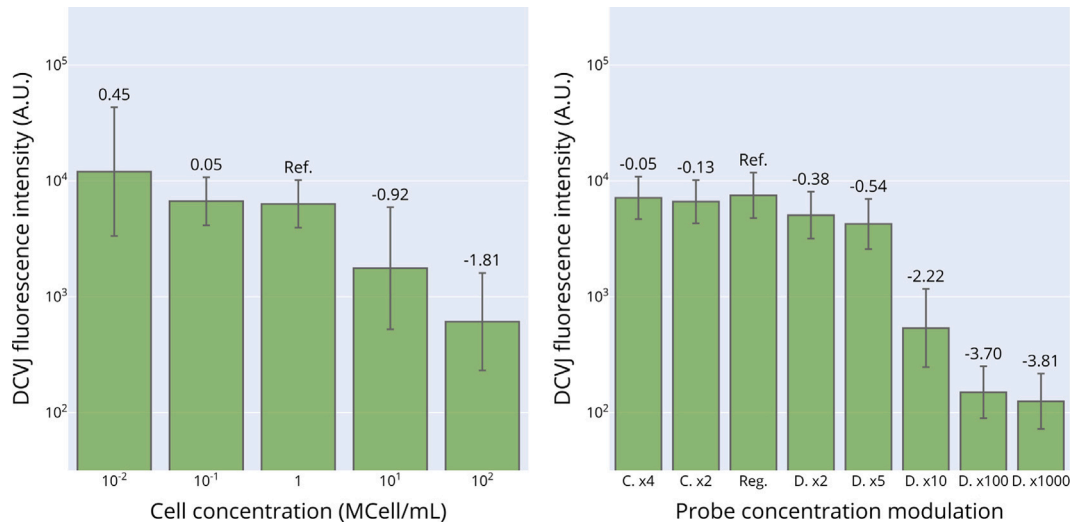


Fig. 2. Left — Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) for different cell concentration (probe volume and concentration — 10 μ L at 10 mM). Right — Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) for different probe concentration (C. — Concentrated, D — Diluted, cell concentration 1 MCell/mL). Sample volume — 1 mL. Incubation time — 20 min in the dark. Error bars — Standard deviation of the population on the DCVJ channel. Values — Cohen's d.

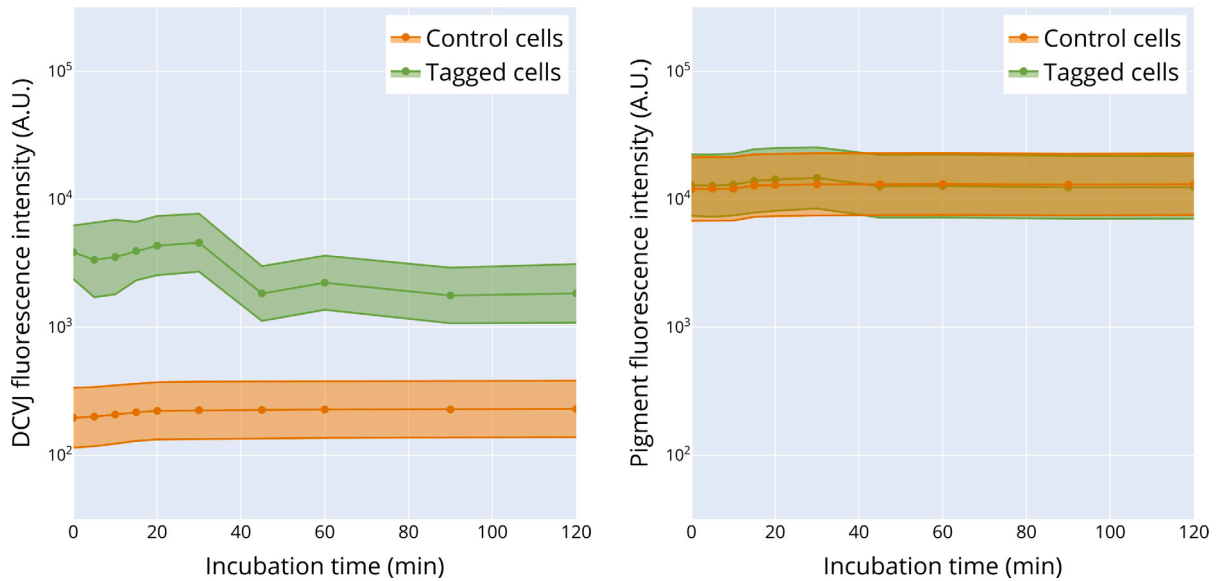


Fig. 3. Left — Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection). Right — Cell fluorescence intensity on the chlorophyll channel (blue laser, 695/40 nm detection). Both graph display varying incubation time, in the dark, for control cells (orange) and tagged cells (green). Shaded area population standard deviation. Sample volume - 1 mL. Cell concentration - 1 MCell/mL. Probe volume and concentration — 10 μ L at 10 mM.

With these elements in mind, a correction procedure can be developed. First, for control (not tagged) cells, the fluorescence on the probe channel (F) is equal to the background (carotenoid) fluorescence (Eq. (3)). For the same cell, the signal on the red channel (P) is equal to the unmodified chlorophyll autofluorescence intensity (Eq. (4)). These two measurements (F and P) represent the reference for signal correction. Moving to the tagged cells, on the red channel, the signal is equal to the chlorophyll autofluorescence signal modified by the membrane alteration (α in Eq. (5)). On the probe channel, the signal is the sum of the background (carotenoids) and probe contributions. Both contributions are modified by the membrane alteration (Eq. (6)). The last step of the correction procedure is to combine these equations. Together, they allow isolating the probe signal of the tagged cells using pigment and probe signals of both control and tagged cells (Eq. (7)). Before applying this correction, it calls for two comments. First, the

alteration of fluorescence signals induced by DCVJ addition is assumed to be multiplicative and shared between pigment and probe channels. Second, the presence of the $-F_{ctl}$ in Eq. (7) underlines the necessity of having a maximal distance between the tagged cell signal and the control cell signal of the probe channel (3 - Left -).

$$F_{ctl} = F_{bg} \quad (3)$$

$$P_{ctl} = P_{bg} \quad (4)$$

$$P_{test} = \alpha P_{bg} \quad (5)$$

$$F_{test} = \alpha(F_{probe} + F_{bg}) \quad (6)$$

$$F_{probe} = F_{test} \frac{P_{ctl}}{P_{test}} - F_{ctl} \quad (7)$$

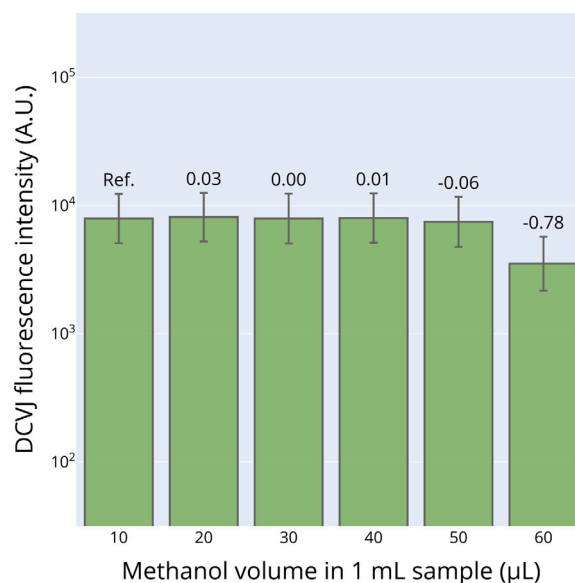


Fig. 4. Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) varying methanol volume introduced in the sample. Sample volume — 1 mL. Cell concentration — 1 MCell/mL. Probe concentration — 10 mM. Incubation time — 20 min, in the dark. Error bars — Standard deviation of the population on the DCVJ channel. Values — Cohen's d.

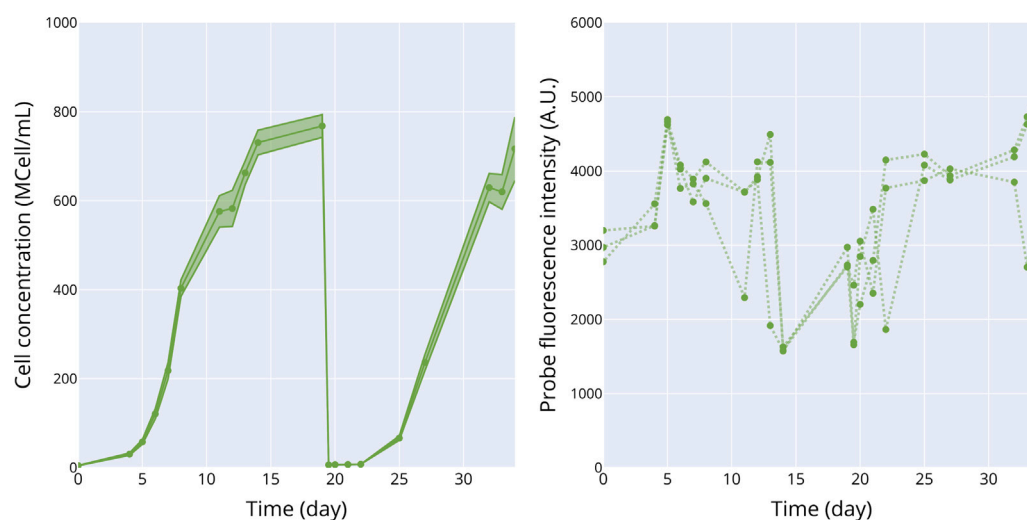


Fig. 5. Left - Cell count over time. Shaded area cell count standard deviation. Right — Cell fluorescence intensity on the DCVJ channel (blue laser, 530/30 nm detection) for small cells (orange) and large cells (green). Sample volume — 1 mL. Cell concentration — 1 MCell/mL. Probe volume and concentration — 10 μL at 10 mM. Incubation time — 20 min, in the dark.

Applying this correction and supplying the obtained signals to Eq. (2), it is now possible to move back to the information of interest: membrane fluidity. Fig. 6 — Right — presents the relative membrane fluidity. As underlined in Eq. (2), going from probe signal to fluidity value required a reference measurement point, which was taken as the one at day 0, sample-wise. Still, as the initial values are extremely close (1281, 1288, and 1299 arbitrary units, after correction), a common reference would have yielded the same results. In addition, the closeness of the corrected values, as opposed to the dispersion of the raw signal (Fig. 5 - Right -), is a token of the quality of the procedure, as these measurements originate from an identical inoculum culture. Focusing on the membrane fluidity trend, one can see that subculturing the cell induced a reduction in cell membrane fluidity. This is the expected outcome as the sharp glucose depletion induces a hyperosmotic shock (from low/regular to hyperosmolar conditions) [35,36]. Then, as the cells proliferate, they consume glucose. Consequently, the

medium osmolarity reduces, and the cells acclimate by fluidizing their membranes. Once glucose is exhausted (early stationary stage, day 19), cells are passaged to a glucose-rich medium, and membranes rigidify once again. All in all, this shows that DCJV membrane tagging allows the tracking of membrane fluidity.

4. Conclusion

This article proposes a flow cytometric methodology to monitor microalgae membrane fluidity. The protocol is based on DCJV, a molecular rotor inserting itself in the cell membrane. After optimization, the best incubation conditions are 1 mL sample at 1 MCell/mL, with a probe addition of 10 μL at 10 mM, incubated in the dark for 20 min. A control sample is also to be analyzed. Advanced signal treatment is required to access membrane fluidity properly. All in all, the protocol was successfully used to monitor microalgae membrane acclimation to hyperosmotic conditions and their reversion.

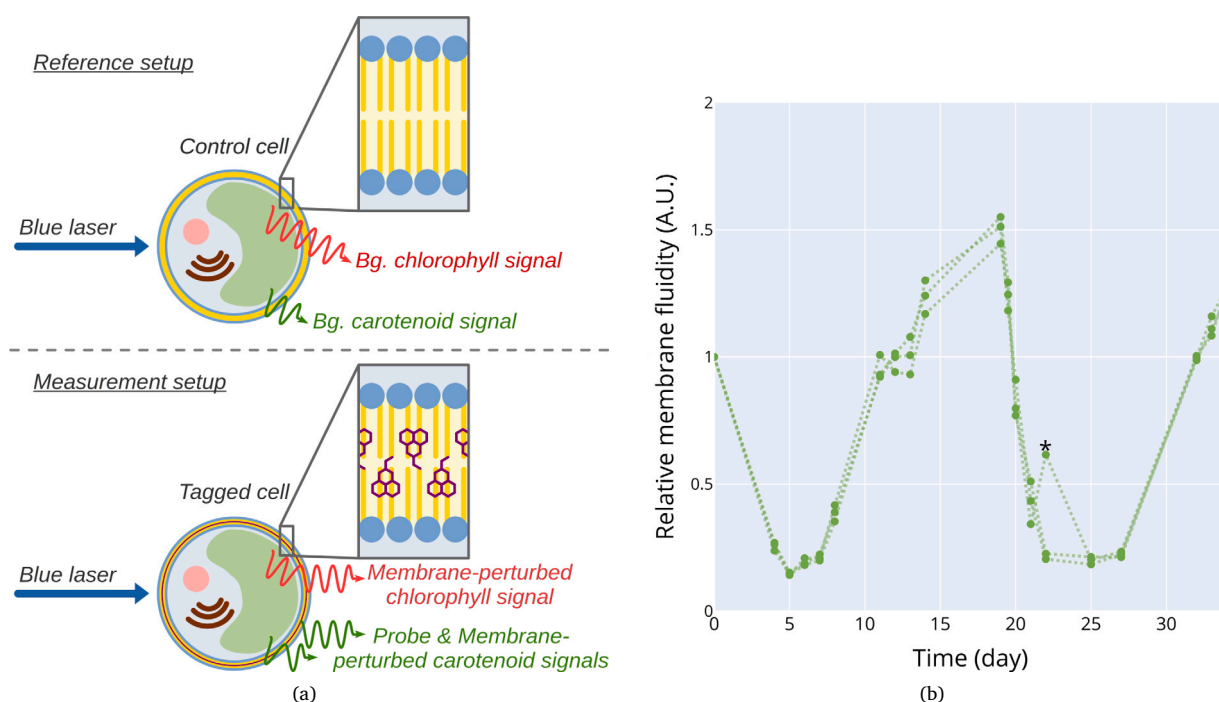


Fig. 6. Left — Signal processing rationale. Right — Cell membrane relative fluidity. Reference value used in Eq. (2) - measurement at day 0, sample-wise. Asterisk — artifact.

CRedit authorship contribution statement

Victor Pozzobon: Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Formal analysis. **Jinjing Yin:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Wendie Levasseur:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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